

MISSION 6

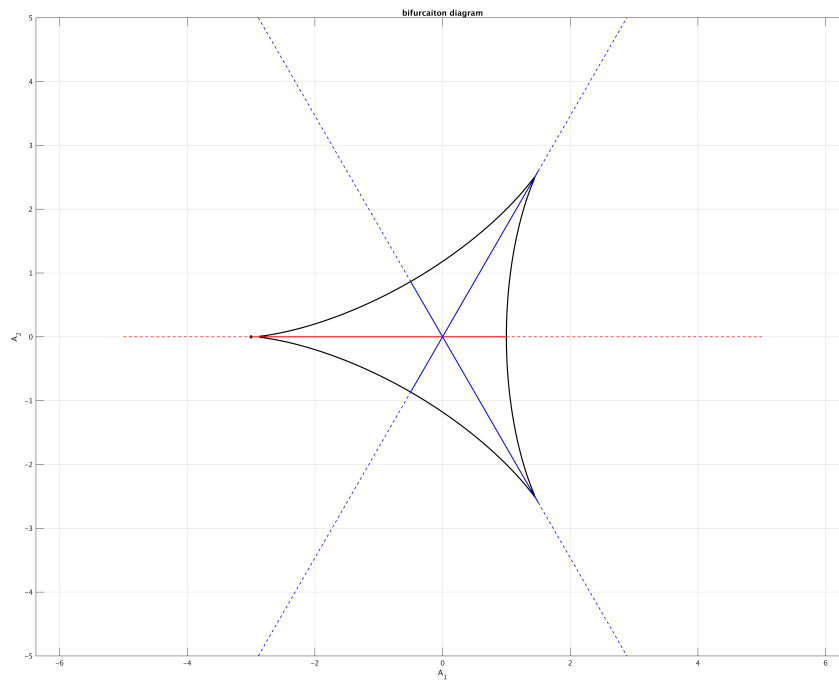
Part 1. Problem

Consider the Hamiltonian system :

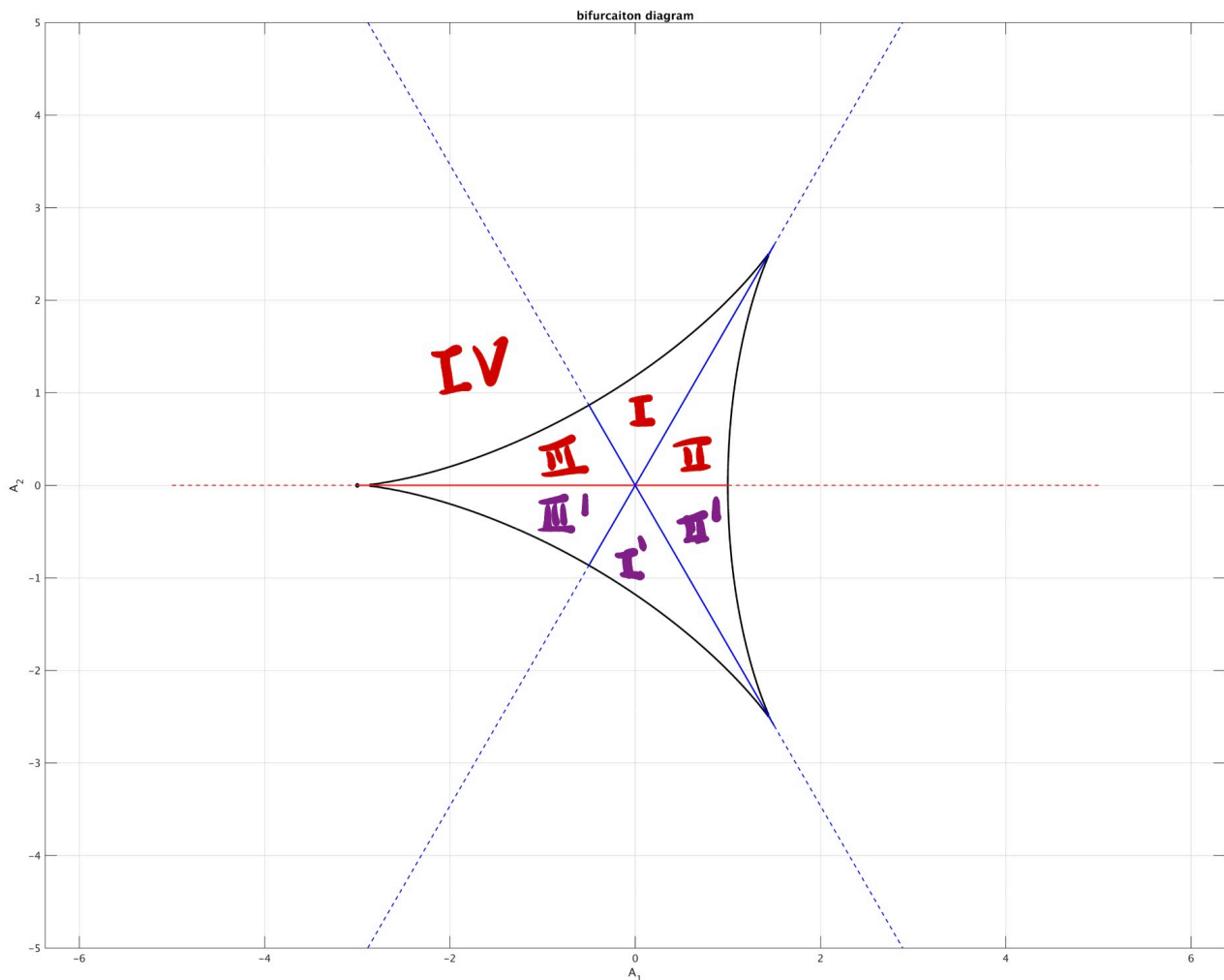
$$H(x, y) = -\frac{1}{2}(x^2 + y^2) - A_2 x^3 - A_2 x y^2 + (1 - A_1) x^2 y - \frac{3A_1 + 1}{3} y^3$$

The bifurcation set is :

$$\begin{cases} \Sigma_1 : & -27 + 18A_1^2 + 8A_1^3 + A_1^4 + 18A_2^2 - 24A_1A_2^2 + 2A_1^2A_2^2 + A_2^4 = 0 \\ \Sigma_2 : & \{A_2 = 0, -3 < A_1 < 1\} \cup \{3A_1^2 - A_2^2 = 0, -\frac{1}{2} < A_1 < \frac{3}{2}\} \end{cases}$$



Problem : determine the phase portrait in each connected region(I,II,III...) of the parameter plane.



Part 2. How to find the phase portrait?

Since for any solution $(\phi(t), \psi(t))$, We have $H(\phi, \psi) = \text{constant}$ (property of Hamiltonian system)

Therefore, Write the phase portrait $\iff$ Write the graph of $H = \text{constant}$.

However, We just need to write some important points is enough (equilibrium points).

Steps to find phase portrait :

Find all Equilibrium points $(x_1, y_1), (x_2, y_2), \dots \implies$ Draw the graph $H(x, y) = H(x_1, y_1), H(x, y) = H(x_2, y_2), \dots$

Part 3. Equilibrium points

The equilibrium equations are $H_x = H_y = 0$:

$$\begin{cases} H_x = -x - 3A_2x^2 - A_2y^2 + 2(1 - A_1)xy = 0 \\ H_y = -y - 2A_2xy + (1 - A_1)x^2 - (3A_1 + 1)y^2 = 0 \end{cases}$$

$(0, 0)$ is always an equilibrium. For $x \neq 0$, set $y = tx$. Eliminating x gives :

$$E(t) := A_2t^3 - (A_1 + 3)t^2 + A_2t + (1 - A_1) = 0$$

Then : solve $t_1, t_2, t_3 \implies$ non-zero equilibrium point $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

Part 4. Find the phase portrait of each regions

Region I:

Take the point $(0, \frac{1}{2}) \in \mathbf{I}$

So :

$$\begin{cases} A_1 = 0 \\ A_2 = \frac{1}{2} \end{cases}$$

Then :

$$\begin{cases} H(x, y) = -\frac{1}{2}(x^2 + y^2) - \frac{1}{2}x^3 - \frac{1}{2}xy^2 + x^2y - \frac{1}{3}y^3 \\ E(t) := t^3 - 6t^2 + t + 2 = 0 \end{cases}$$

$\Rightarrow$

$$\begin{cases} t_1 \approx -0.48361162 \\ t_2 \approx 0.717176 \\ t_3 \approx 5.766435 \end{cases}$$

$\Rightarrow$

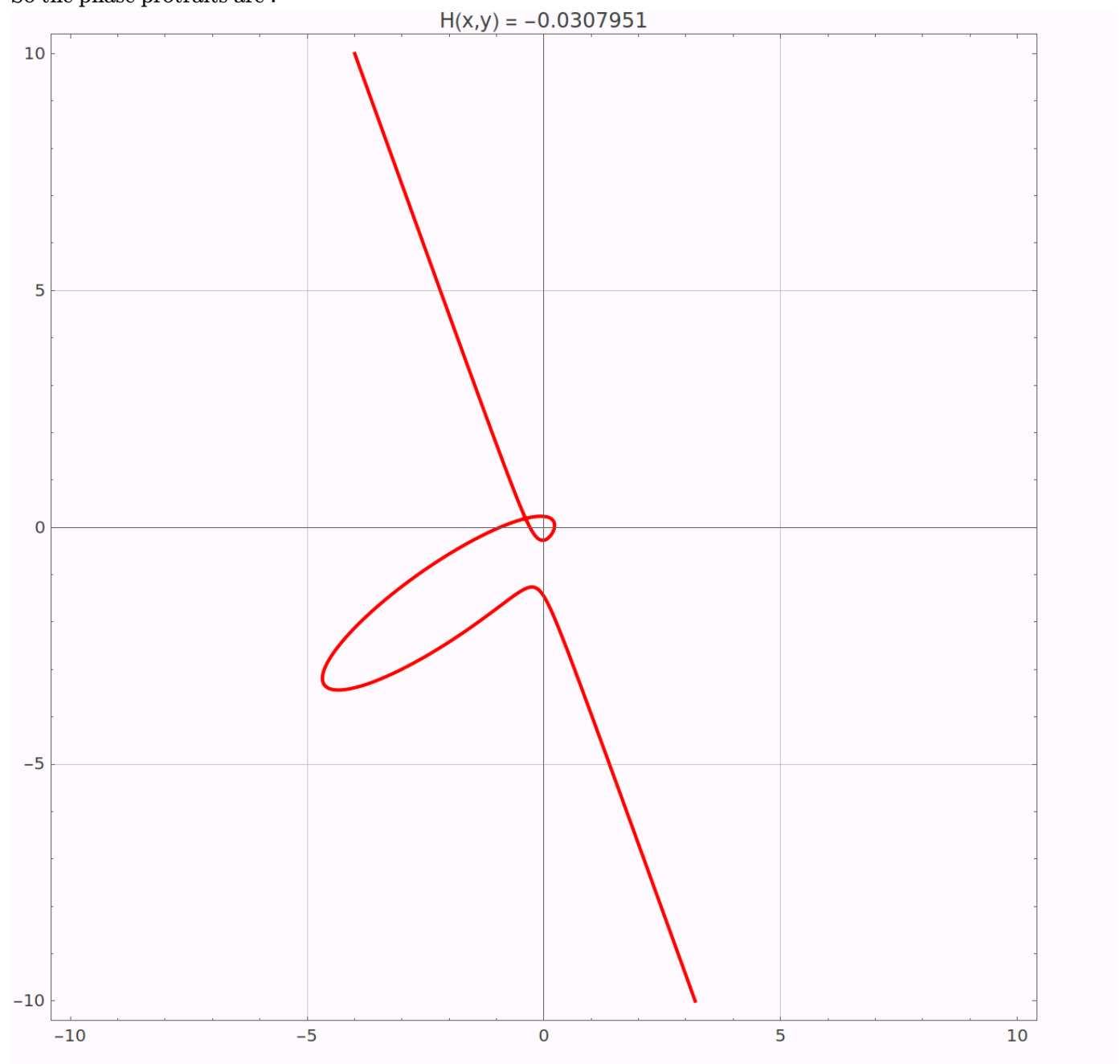
Equilibrium points are :

$$\begin{cases} (x_1, y_1) = (-0.386972, 0.187144) \\ (x_2, y_2) = (-3.09771, -2.22161) \\ (x_3, y_3) = (-0.151676, -0.874628) \\ (x_4, y_4) = (0, 0) \end{cases}$$

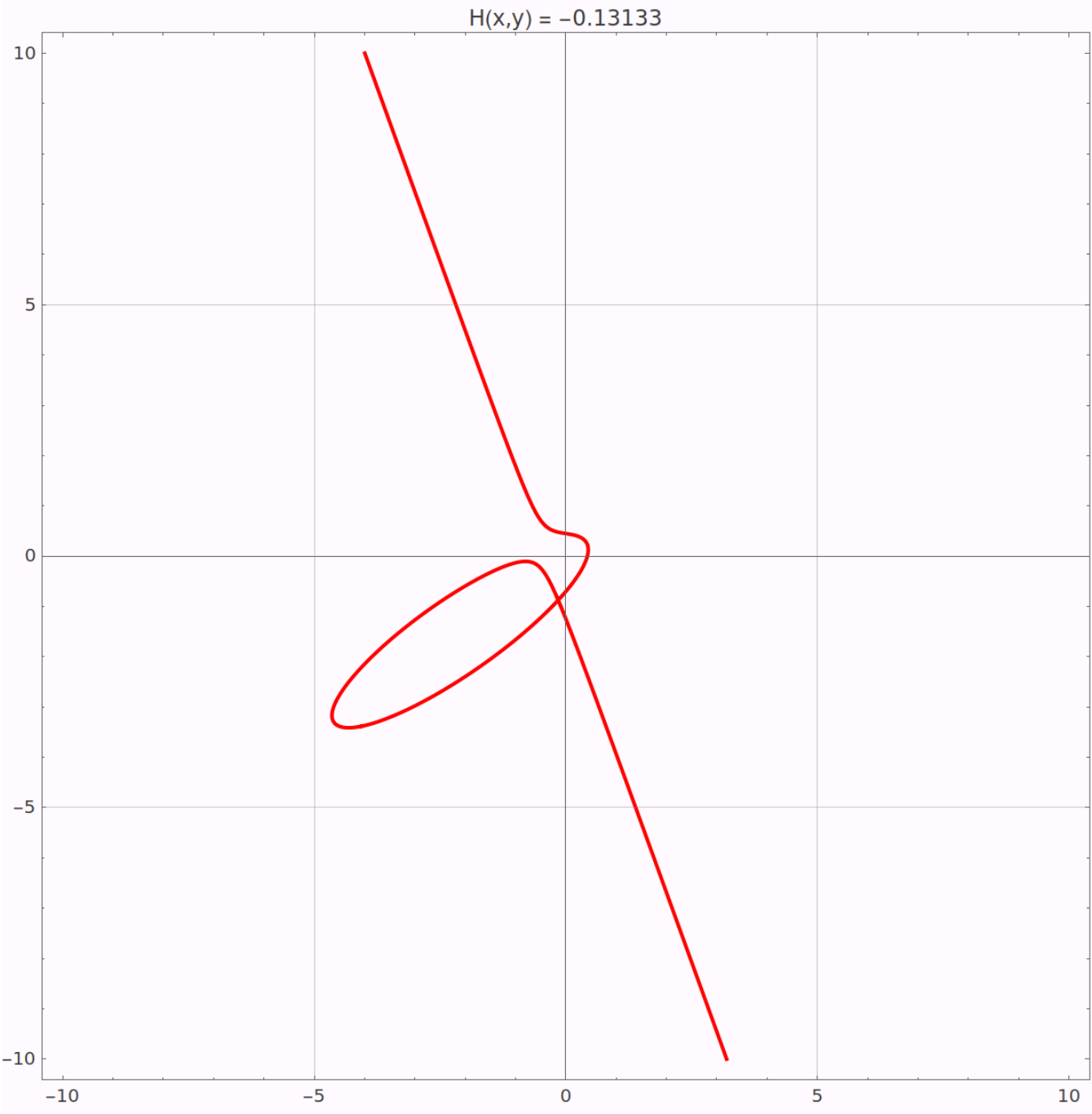
$\Rightarrow$

$$\begin{cases} H(x_1, y_1) = -0.0307951 \\ H(x_2, y_2) = -2.4219 \\ H(x_3, y_3) = -0.13133 \\ H(x_4, y_4) = 0 \end{cases}$$

So the phase portraits are :









Region II :

Take the point $(\frac{1}{2}, \frac{1}{2}) \in \text{II}$

So :

$$\begin{cases} A_1 = \frac{1}{2} \\ A_2 = \frac{1}{2} \end{cases}$$

Then :

$$\begin{cases} H(x,y) = -\frac{1}{2}(x^2 + y^2) - \frac{1}{2}x^3 - \frac{1}{2}xy^2 + \frac{1}{2}x^2y - \frac{5}{6}y^3 \\ E(t) := \frac{1}{2}t^3 - \frac{7}{2}t^2 + \frac{1}{2}t + \frac{1}{2} = 0 \end{cases}$$

$\implies$

$$\begin{cases} t_1 \approx -7.1207132 \\ t_2 \approx -0.3192198 \\ t_3 \approx 0.4399331 \end{cases}$$

⇒

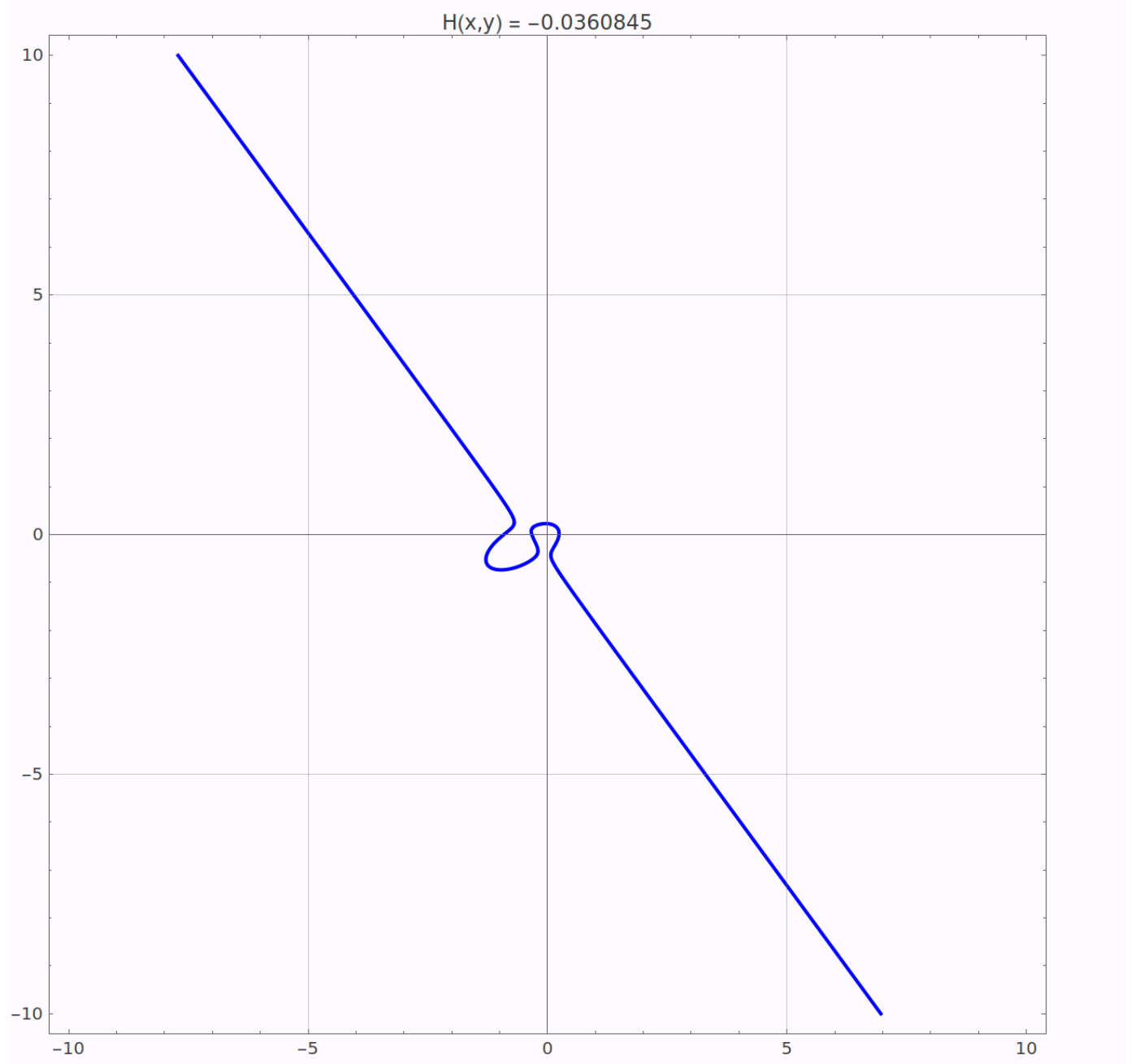
Equilibrium points are :

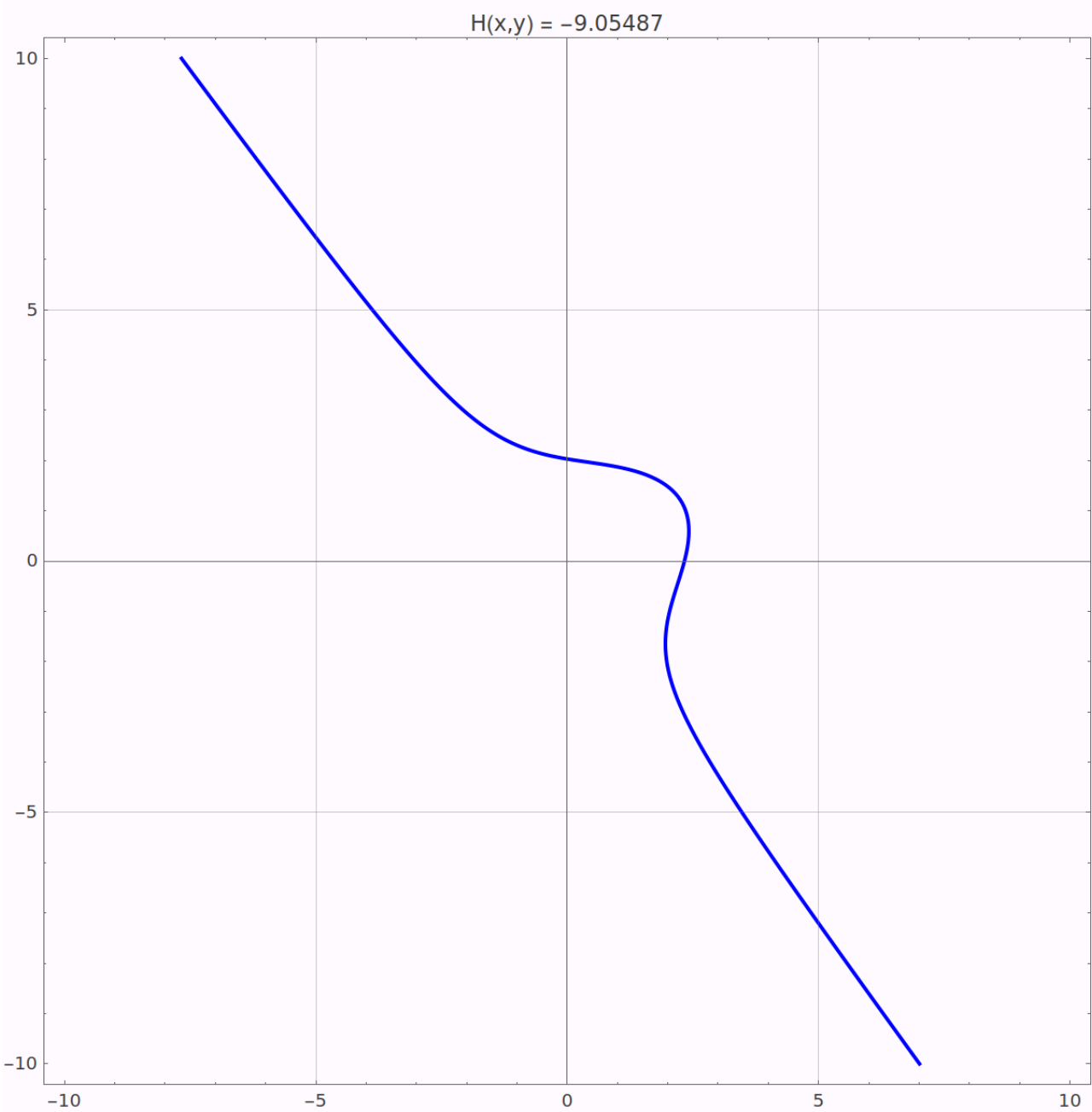
$$\begin{cases} (x_1, y_1) = (-0.0322861, 0.2299007) \\ (x_2, y_2) = (0.8850892, -0.28253801) \\ (x_3, y_3) = (1.03810215, -0.45669549) \\ (x_4, y_4) = (0, 0) \end{cases}$$

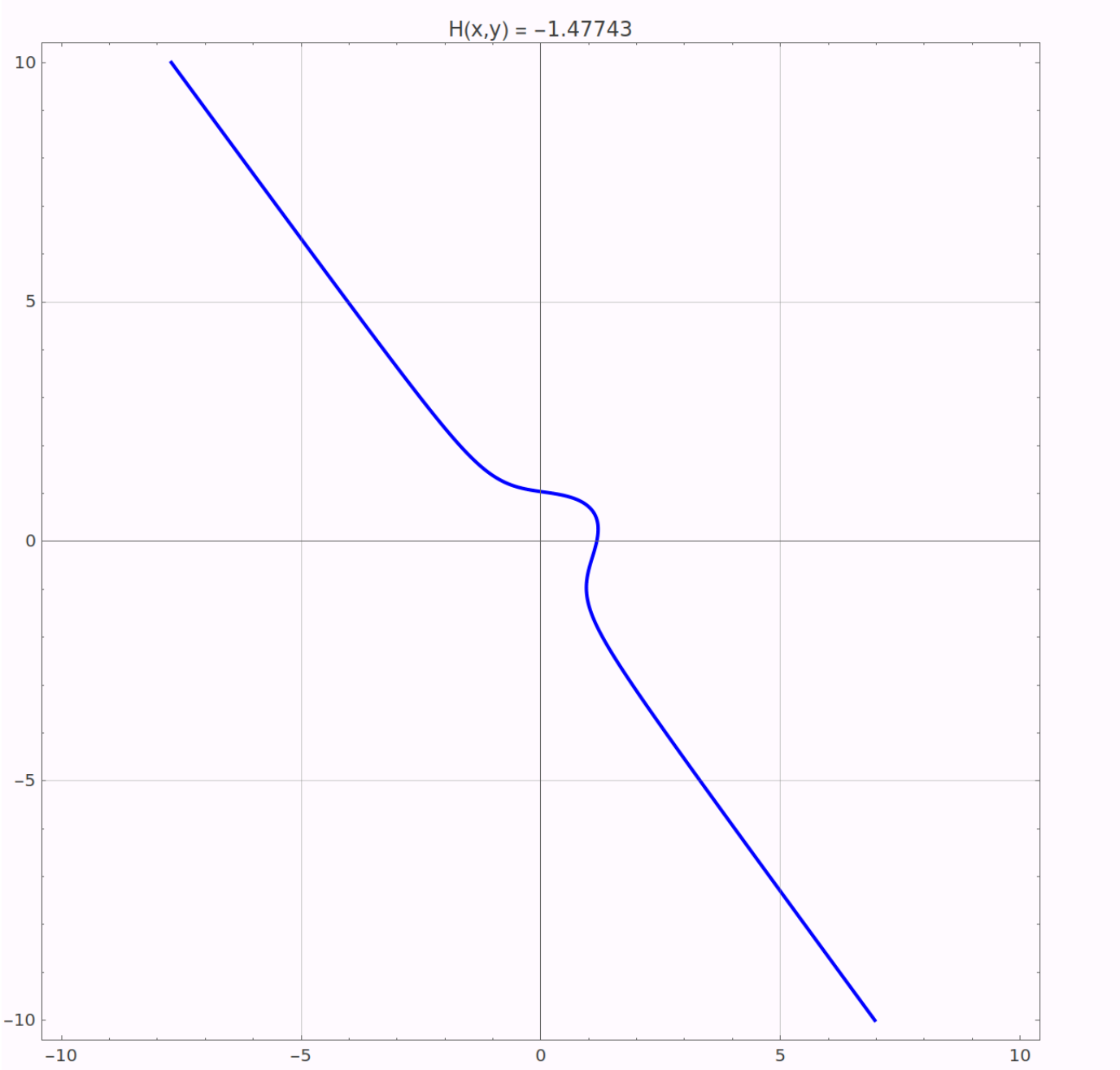
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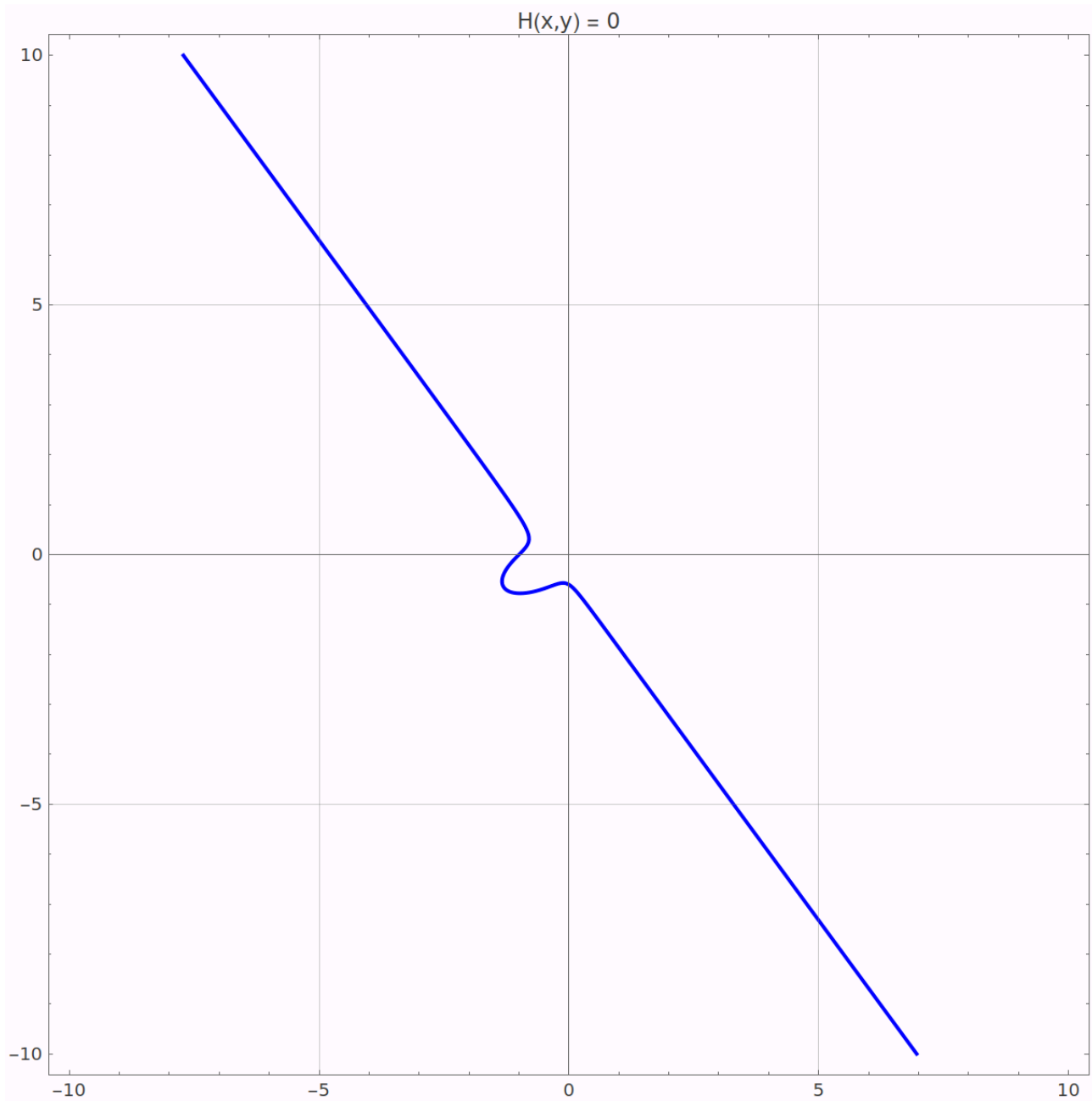
$$\begin{cases} H(x_1, y_1) = -0.036084521 \\ H(x_2, y_2) = -9.0548692 \\ H(x_3, y_3) = -1.4774334 \\ H(x_4, y_4) = 0 \end{cases}$$

So the phase portraits are :









Region III :

Take the point $(-1, \frac{1}{2}) \in \text{III}$

So :

$$\begin{cases} A_1 = -1 \\ A_2 = \frac{1}{2} \end{cases}$$

Then :

$$\begin{cases} H(x, y) = -\frac{1}{2}(x^2 + y^2) - \frac{1}{2}x^3 - \frac{1}{2}xy^2 + 2x^2y + \frac{2}{3}y^3 \\ E(t) := \frac{1}{2}t^3 - 2t^2 + \frac{1}{2}t + 2 = 0 \end{cases}$$

$\implies$

$$\begin{cases} t_1 \approx -0.81360650 \\ t_2 \approx 1.470683419 \\ t_3 \approx 3.342923082 \end{cases}$$

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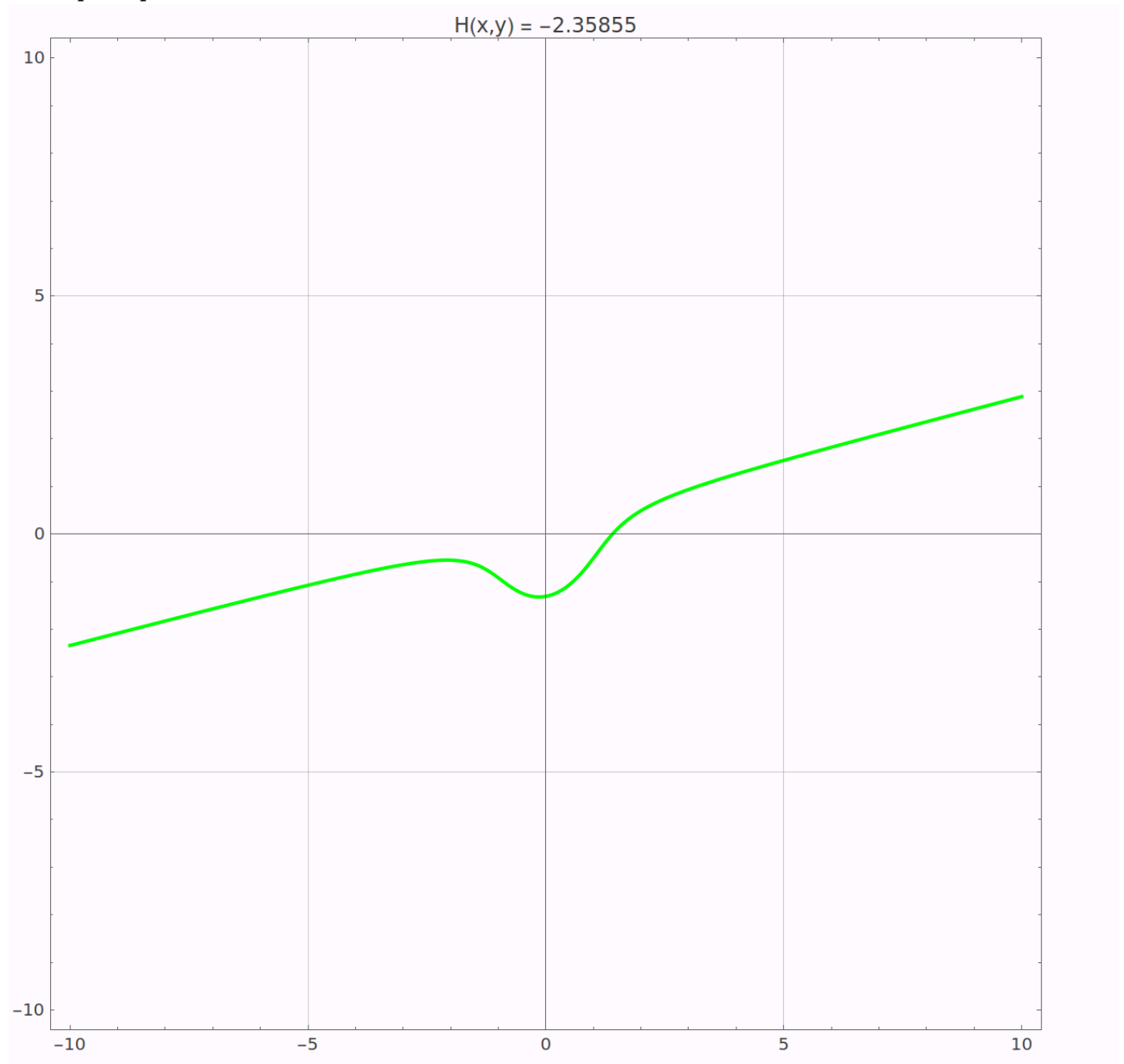
Equilibrium points are :

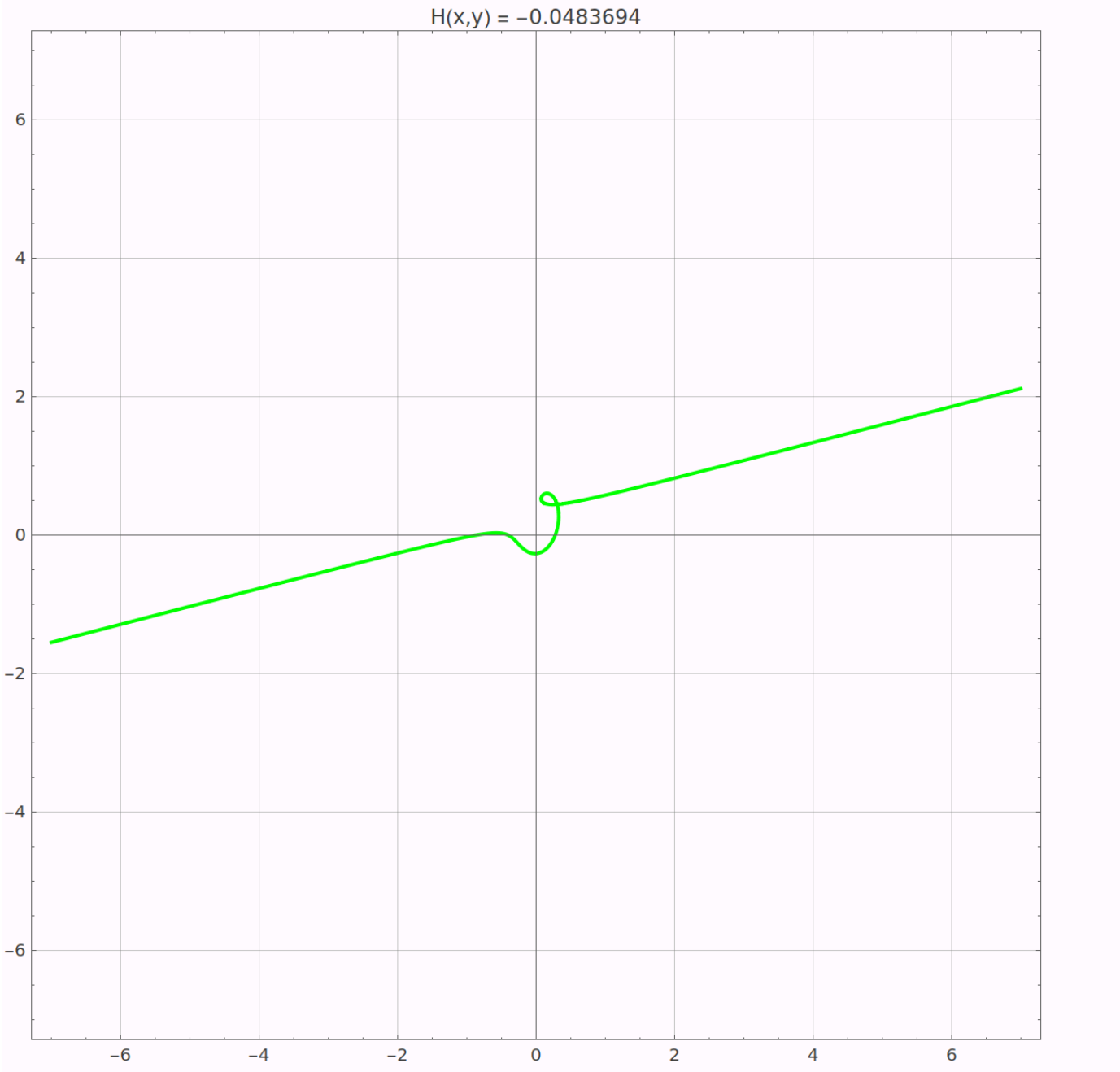
$$\begin{cases} (x_1, y_1) = (-0.19664121, 0.159988574) \\ (x_2, y_2) = (0.3029129181, 0.445489006) \\ (x_3, y_3) = (0.15913114491, 0.53196317741) \\ (x_4, y_4) = (0, 0) \end{cases}$$

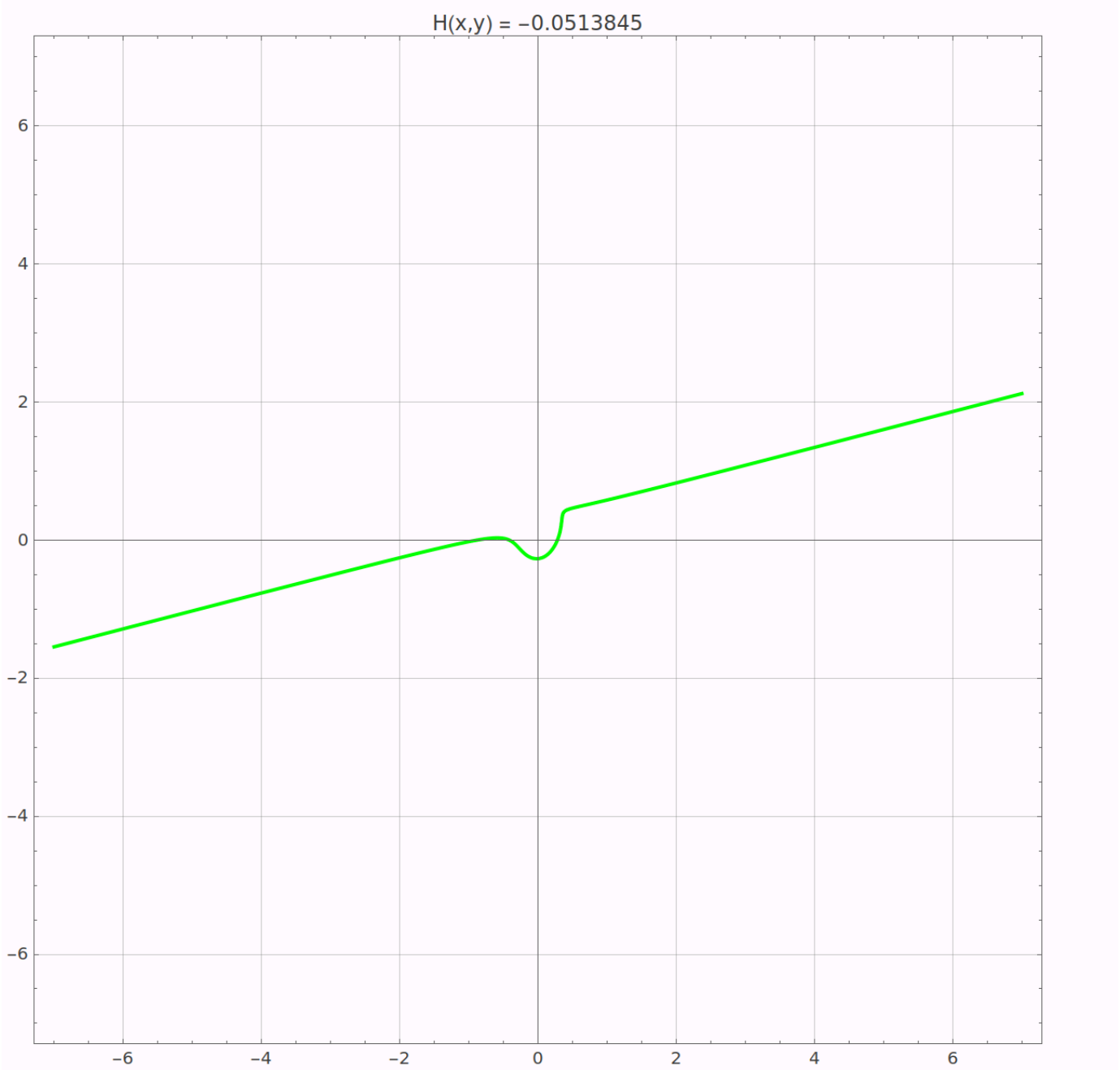
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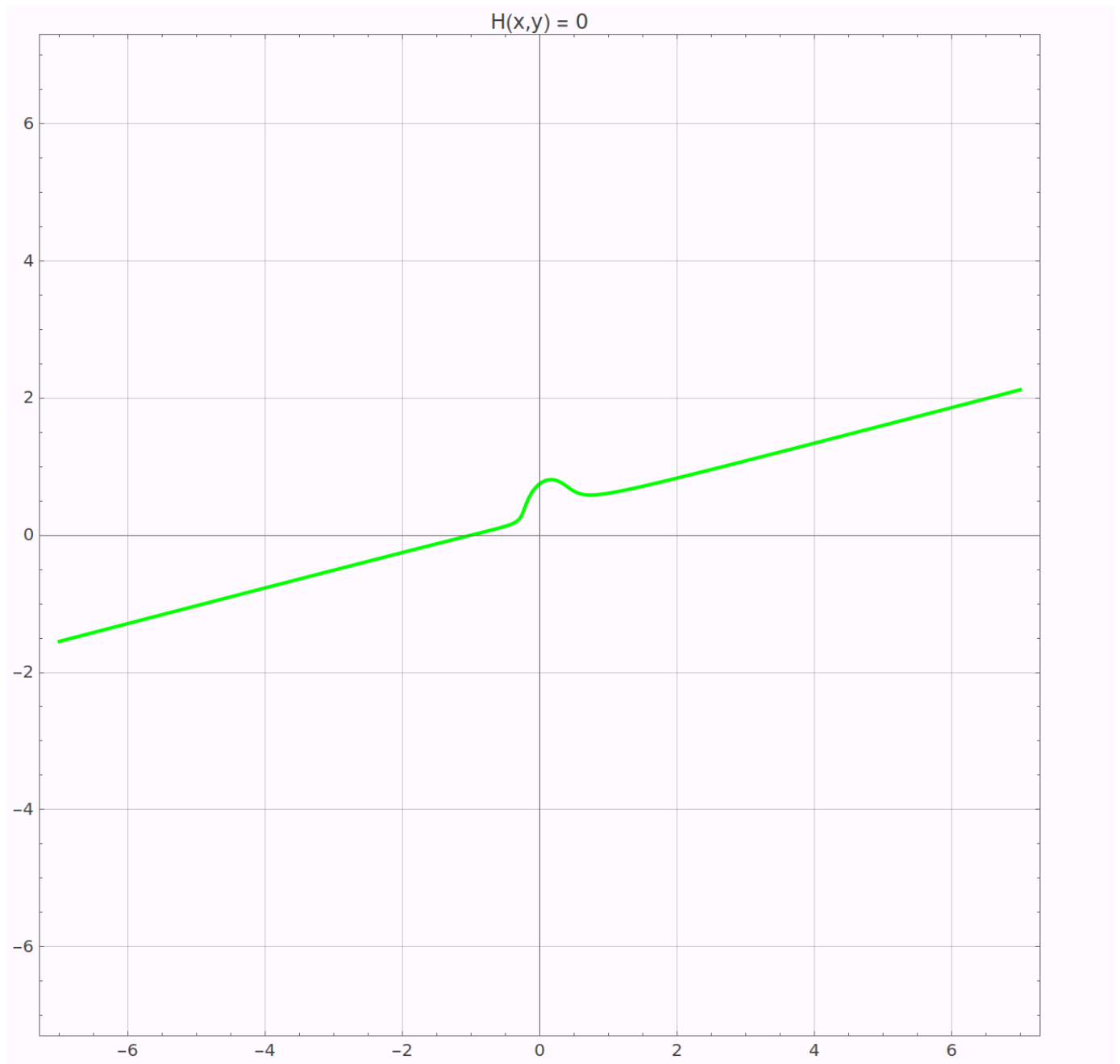
$$\begin{cases} H(x_1, y_1) = -2.35855440 \\ H(x_2, y_2) = -0.0483694484 \\ H(x_3, y_3) = -0.0513845 \\ H(x_4, y_4) = 0 \end{cases}$$

So the phase portraits are :









Region IV :

Take the point $(-4, 2) \in \text{IV}$

So :

$$\begin{cases} A_1 = -4 \\ A_2 = 2 \end{cases}$$

Then :

$$\begin{cases} H(x,y) = -\frac{1}{2}(x^2 + y^2) - 2x^3 - 2xy^2 + 5x^2y + \frac{11}{3}y^3 \\ E(t) := 2t^3 + t^2 + 2t + 5 = 0 \end{cases}$$

$\Rightarrow$

$$\{t_1 \approx -1.26738313$$

$\Rightarrow$

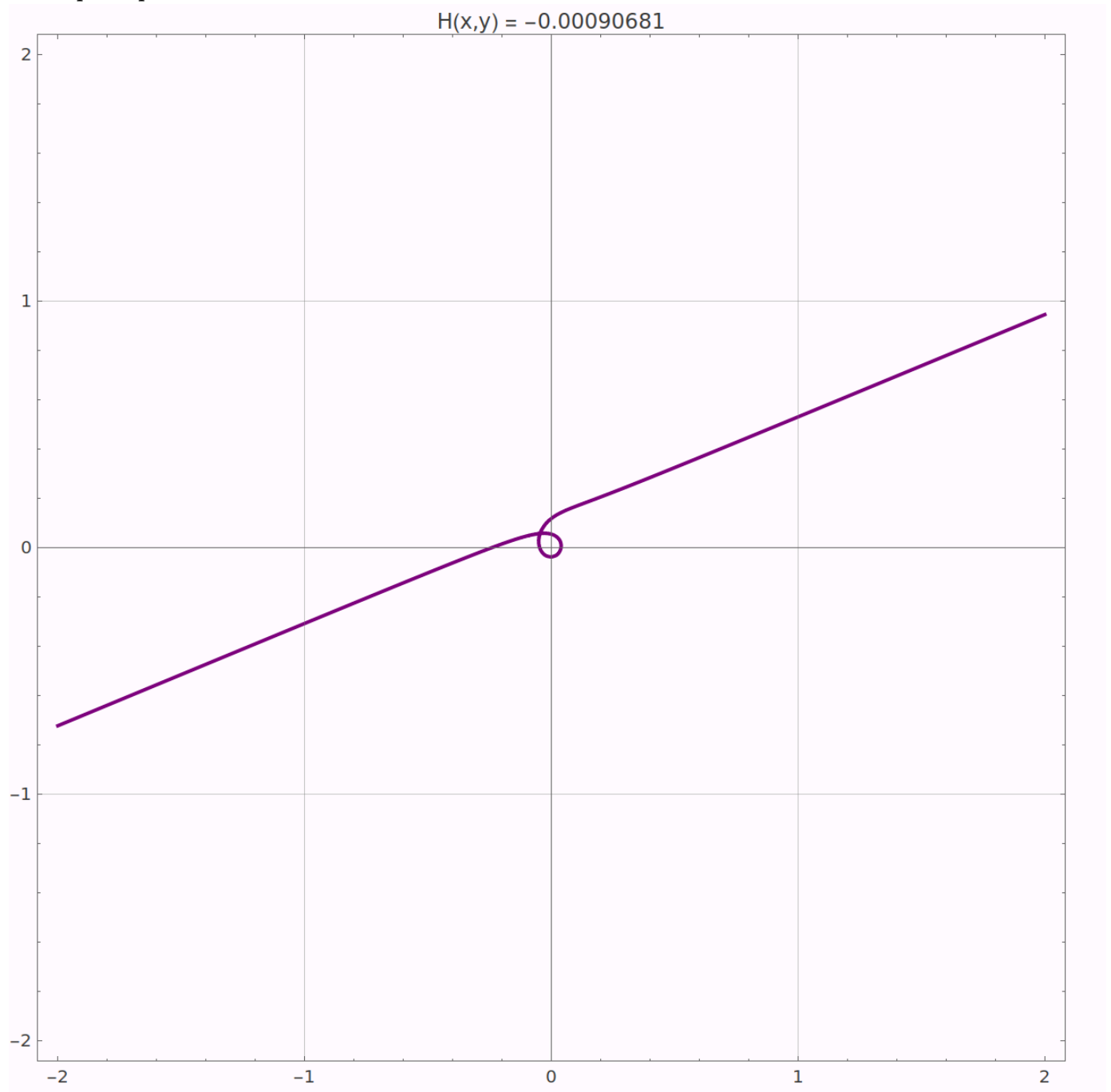
Equilibrium points are :

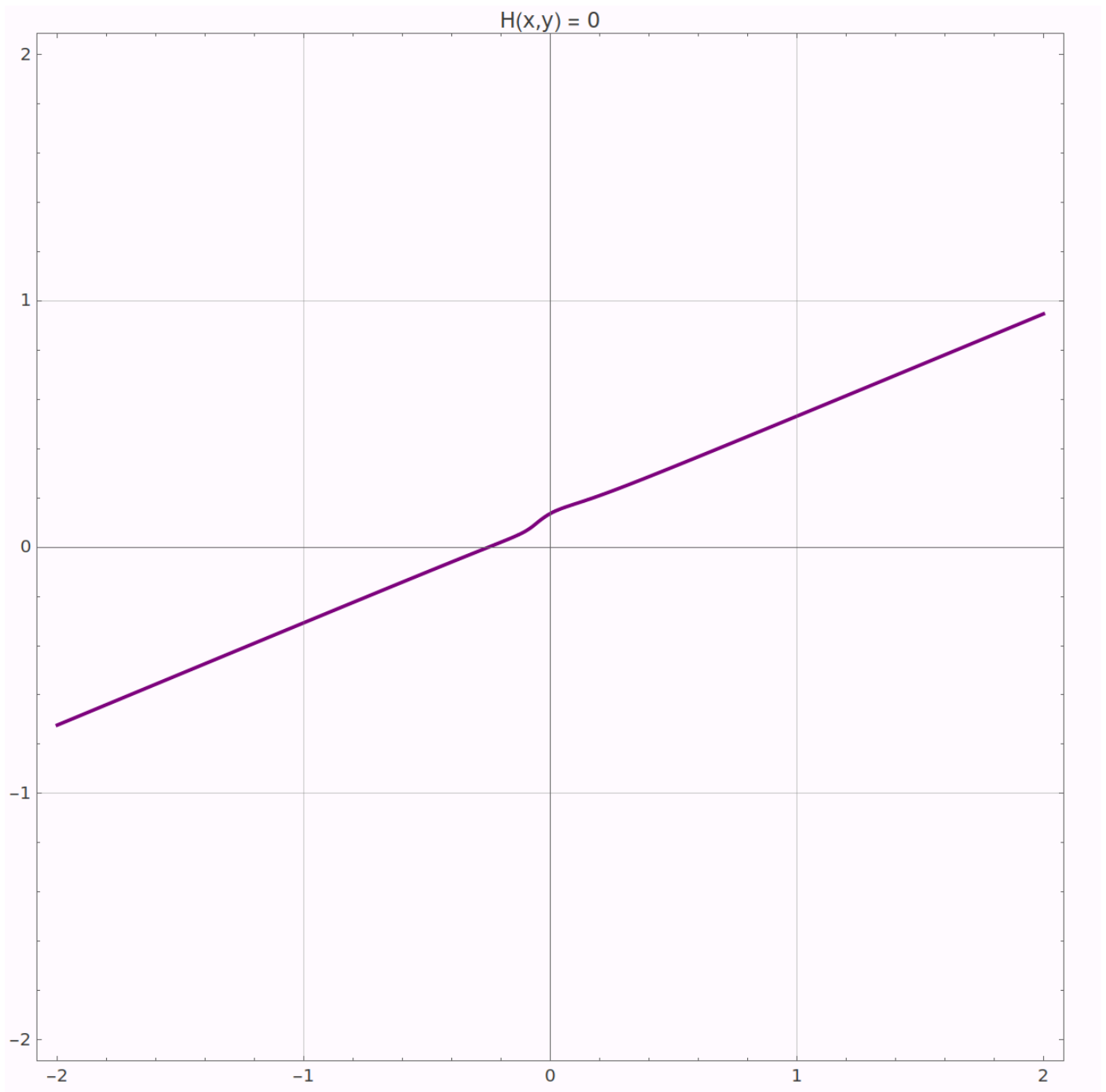
$$\begin{cases} (x_1, y_1) = (-0.04569057, 0.057907465) \\ (x_2, y_2) = (0, 0) \end{cases}$$

$\Rightarrow$

$$\begin{cases} H(x_1, y_1) = -0.00090681 \\ H(x_2, y_2) = 0 \end{cases}$$

So the phase portraits are :





Part 5. Remarks

since the symmetric of $H(x,y)$ and Σ_1

We have :

$$H_{A_1, -A_2}(x, y) = H_{A_1, A_2}(-x, y)$$

This means We just need to solve regions **I,II,III,IV**

I',II',III' are symmetric.